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Implementation of Machine Learning project

Customer Sentiment Analysis on Social Media Posts using Machine Learning

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Contents

Introduction **2**

1 Methodology **3**

1.1 Overview . . . 3

1.2 Steps in the Methodology . . . 3

1.2.1 Data Loading and Exploration . . . 3

1.2.2 Data Preprocessing . . . 3

1.2.3 Feature Extraction . . . 4

1.2.4 Handling Class Imbalance . . . 4

1.2.5 Model Training and Evaluation . . . 5

1.2.6 Evaluation Metrics . . . 5

1.2.7 Implementation . . . 5

1.3 Conclusion . . . 6

2 Data Analysis **7**

2.1 Dataset Overview . . . 7

2.1.1 Dataset Description . . . 7

2.1.1.1 Key Features . . . 7

2.1.1.2 Why this Dataset? . . . 8

2.1.2 Exploratory Data Analysis (EDA) . . . 8

2.2 Dataset Diagnostics . . . 10

2.3 Data Preprocessing . . . 10

2.4 Insights from Data Analysis . . . 11

2.5 Sample Records . . . 11

2.6 Conclusion . . . 11

3 Results and Discussion **12**

3.1 Data Splitting . . . 12

3.2 Model Training and Performance 12
3.2.1 Support Vector Machine (SVM) 12
3.2.2 KNN 13
3.2.3 Naive Bayes 15
3.2.4 Decision Tree 16
3.2.5 Random Forest 17
3.3 Performance Summary 18
3.4 Discussion 18

Conclusion **19**

List of Figures

1.1 Number of sentiments of tweets for each class 4

2.1 Histogram of tweet length 9
2.2 Reasons of negative tweets 10

3.1 Classification report for the SVM model 13
3.2 Visualization of the SVM model's ROC curve and confusion matrix 13
3.3 Classification report for the SVM model 14
3.4 Confusion matrix off KNN model 14
3.5 Classification report for the Naive Bayes model 15
3.6 Visualization of the Naive Bayes model's ROC curve and confusion matrix 15
3.7 Classification report for the Decision Tree model 16
3.8 Visualization of the Decision Tree model's ROC curve and confusion matrix 16
3.9 Classification report for the Random Forest model 17
3.10 Visualization of the Random Forest model's ROC curve and confusion matrix 17

Abstract

The rise of social media platforms like Twitter and Facebook has provided vast amounts of user-generated data. Sentiment analysis leverages machine learning and natural language processing (NLP) to extract insights from these data by identifying emotions, opinions, and attitudes. This project focuses on analyzing customer sentiment from social media posts using techniques such as SVM, KNN and Random Forest . By classifying posts as positive (2) , neutral (1) or negative (0), businesses can track customer satisfaction trends and adapt strategies effectively. This study presents a systematic approach to customer sentiment analysis, combining computational methods with practical business applications.

Introduction

In today's digital age, social media platforms have become key mediums for people to share experiences, express opinions, and discuss ideas. Platforms such as Twitter and Facebook host vast amounts of emotional and opinion-based content, offering unique opportunities for analysis. Understanding this user-generated data is essential for businesses aiming to stay competitive and responsive to customer needs.

Sentiment analysis, a subset of natural language processing (NLP), focuses on identifying and categorizing opinions expressed in text. It evaluates whether a piece of content conveys a positive, negative, or neutral sentiment. This technology provides actionable insights into customer feedback, enabling companies to monitor brand perception, track customer satisfaction, and improve decision-making processes.

The goal of this project is to design a sentiment analysis system using machine learning algorithms, including Support Vector Machines (SVM), K nearest neighbors (KNN), naive bayes, decison tree and Random Forest . By leveraging these methods, we aim to extract meaningful patterns from unstructured social media data. This project stands out by its comprehensive approach, integrating multiple models to achieve reliable sentiment classification. Furthermore, it aligns with real-world business scenarios, offering valuable insights into customer preferences and trends.

Through this study, we aim to explore the potential of machine learning in understanding social media sentiments, bridging the gap between raw data and actionable insights. The findings of this work highlight the importance of automated sentiment analysis in a data-driven world, where businesses must adapt swiftly to customer demands and public opinions.

Chapter 1

Methodology

1.1 Overview

This chapter outlines the structured methodology employed for sentiment analysis on customer tweets. The approach incorporates data preprocessing, feature extraction, model building, and evaluation to ensure robust results.

1.2 Steps in the Methodology

1.2.1 Data Loading and Exploration

The first step involves loading the dataset and performing an exploratory data analysis (EDA). This includes:

- Inspecting the dataset structure using summary statistics and visualizations.
- Analyzing the distribution of sentiments (positive, neutral, negative) and other relevant categorical variables.
- Exploring airline-specific sentiment distributions to uncover patterns in the data.

1.2.2 Data Preprocessing

To prepare the text data for analysis, the following preprocessing steps were performed:

- **Noise Removal:** Elimination of special characters, punctuation, URLs, and usernames.

- **Stopword Removal:** Excluding common words that do not contribute meaningfully to sentiment.
- **Text Normalization:** Converting text to lowercase and standardizing contractions.
- **Decontraction:** Reducing words to their base forms for consistent representation.

1.2.3 Feature Extraction

The preprocessed text was transformed into numerical features suitable for machine learning using:

- **TF-IDF (Term Frequency-Inverse Document Frequency):** Captures the importance of terms relative to the corpus.

1.2.4 Handling Class Imbalance

We notice we have a severe class imbalance with the majority class being the negative one as shown in the following plot:

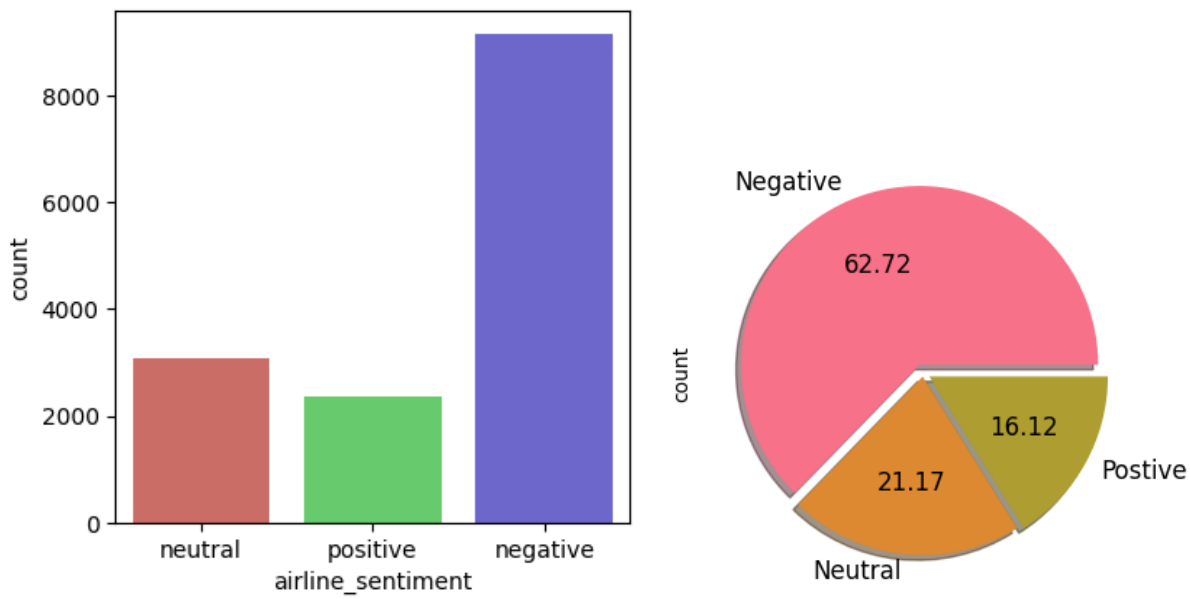

Figure 1.1: Number of sentiments of tweets for each class

That's why, **SMOTE (Synthetic Minority Over-sampling Technique)** was employed to generate synthetic examples in underrepresented classes, ensuring a balanced dataset for training.

1.2.5 Model Training and Evaluation

The following machine learning algorithms were implemented to classify sentiments:

- **Support Vector Machine (SVM):** Known for its effectiveness in high-dimensional spaces.
- **K-Nearest Neighbors (KNN):** A simple, instance-based learning algorithm.
- **Naive Bayes:** A probabilistic classifier suitable for text data.
- **Decision Tree:** A non-parametric algorithm that models hierarchical splits.
- **Random Forest:** An ensemble method combining multiple decision trees to improve accuracy.

1.2.6 Evaluation Metrics

To assess the model performance, the following metrics were employed:

- **Accuracy:** Percentage of correct predictions.
- **Precision:** Measure of exactness in positive predictions.
- **Recall:** Measure of sensitivity to actual positive cases.
- **F1-Score:** Harmonic mean of precision and recall, balancing both metrics.

1.2.7 Implementation

The implementation steps included:

- Applying the preprocessing pipeline to the dataset.
- Splitting the cleaned dataset into training and testing subsets.
- Training and comparing machine learning models.
- Selecting the best-performing model based on evaluation metrics.

1.3 Conclusion

This chapter described the systematic methodology for sentiment analysis. By employing advanced preprocessing techniques, robust feature extraction, and multiple classifiers, the methodology ensures accurate and reliable insights into customer sentiments.

Chapter 2

Data Analysis

2.1 Dataset Overview

The Twitter US Airline Sentiment dataset is a widely-used dataset in sentiment analysis and natural language processing (NLP). It was initially released as part of a Kaggle competition and contains data collected from Twitter to analyze customer sentiments about US airlines.

2.1.1 Dataset Description

2.1.1.1 Key Features

1. **Source:**
   - Data is derived from tweets related to major US airlines.
   - Includes both customer complaints and positive experiences.
2. **Size:**
   - Contains 14,640 labeled tweets.
3. **Sentiment Classes:**
   - **Positive:** Indicates a positive sentiment towards the airline.
   - **Neutral:** Indicates a neutral sentiment or statement of fact.
   - **Negative:** Indicates dissatisfaction or complaints about the airline.

2.1.1.2 Why this Dataset?

The Twitter US Airline Sentiment Dataset is an excellent choice for sentiment analysis projects due to several reasons:

1. **Well-Labeled**
   - The dataset is already annotated with sentiment labels (positive, neutral, negative) and includes reasons for negative sentiments.
   - It simplifies the task of supervised learning since you don't need to label data manually.
2. **Relevant to Real-World Applications**
   - **Customer Feedback:** Airlines often rely on social media for customer interactions. This dataset is directly tied to understanding customer satisfaction, complaints, and service efficiency.
   - **Brand Reputation Management:** Insights from this dataset can guide marketing strategies and service improvements.

2.1.2 Exploratory Data Analysis (EDA)

To gain insights into the dataset, several diagnostic tools and visualizations were employed:

- **Sentiment Distribution:** The dataset exhibits a significant class imbalance, with negative sentiments dominating. This imbalance highlights the need for resampling techniques during model training.
- **Text Length Analysis:** The length of tweets varies widely, with an average of 103 characters. A histogram of tweet lengths reveals that the minimum and maximum length are respectively 12 and 186. However, one can say that most tweets fall within 20–150 characters as shown in the following histogram plot:

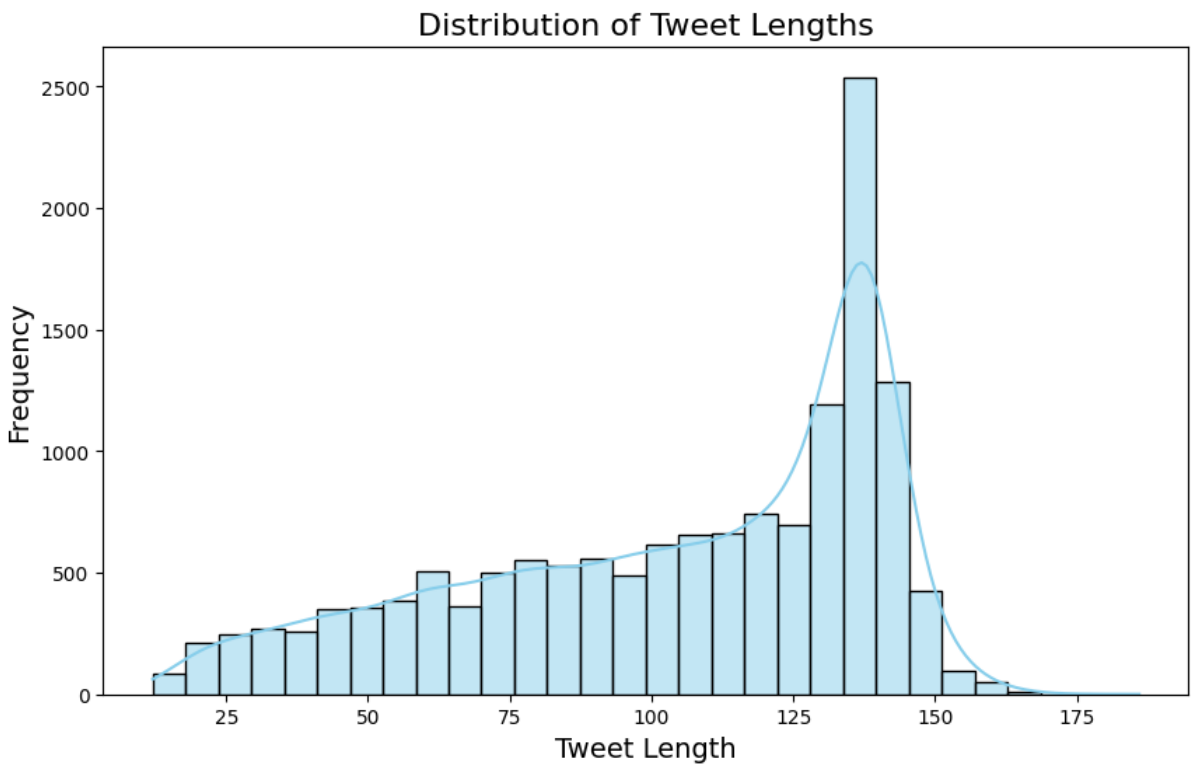

Figure 2.1: Histogram of tweet length

- **Reasons of negative tweets:** The dataset highlights Customer Service Issues as the leading cause of negative tweets, followed by Late Flights and Cancelled Flights. These predominant categories underscore the importance of addressing operational inefficiencies to enhance customer satisfaction. Figure 3.1 summarizes the most common reasons for dissatisfaction.

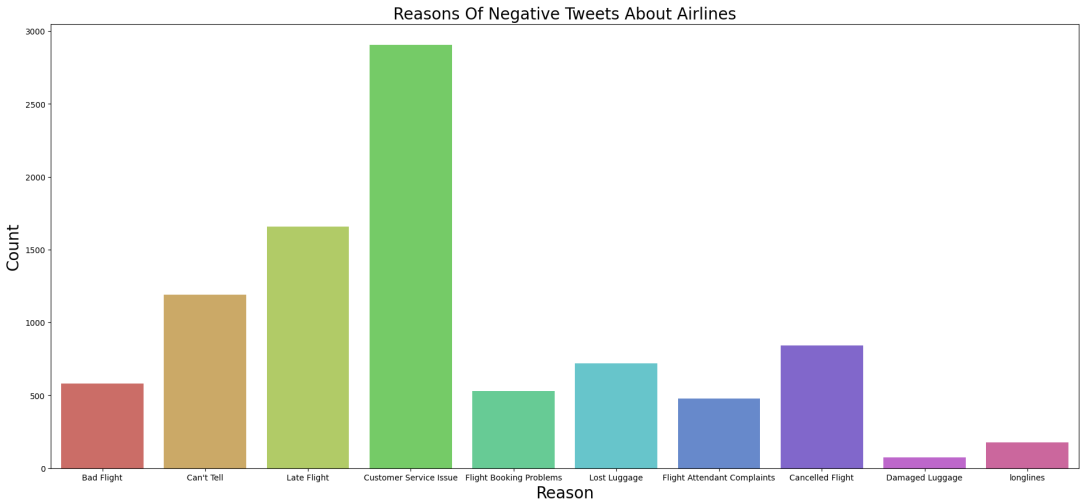

Figure 2.2: Reasons of negative tweets

2.2 Dataset Diagnostics

To ensure data quality and reliability, the following diagnostics were conducted:

- **Missing Values:** No missing values were found in the dataset.
- **Duplicate Records:** 36 duplicate tweets were identified and removed to avoid bias in the analysis.

2.3 Data Preprocessing

Preprocessing was critical to transforming raw tweets into a clean format suitable for analysis. The following steps were applied:

- **Noise Removal:** URLs, usernames (@mentions), hashtags, and special characters were removed.
- **Case Normalization:** All text was converted to lowercase for uniformity.
- **Stopword Removal:** Non-informative words (e.g., "and," "the") were eliminated using the NLTK library.
- **Lemmatization:** Words were reduced to their base forms to minimize redundancy (e.g., "delayed" → "delay").

- **Handling Class Imbalance:** SMOTE (Synthetic Minority Over-sampling Technique) was employed to balance sentiment classes.

2.4 Insights from Data Analysis

- **Class Imbalance:** The predominance of negative sentiments reflects a bias toward customer complaints in the dataset. Addressing this imbalance is crucial for training unbiased models.
- **Dataset Richness:** The diversity in tweet content and length indicates that the dataset effectively captures a wide range of customer experiences.

2.5 Sample Records

Below are examples of tweets from each sentiment class (we opted to show the real ones without removal discussed in the data preprocessing section):

- **Positive:** "@VirginAmerica you know what would be amazingly awesome? BOS-FLL PLEASE!!!!!!! I want to fly with only you."
- **Negative:** "@united every time I search a flight your site logs me out and gives error message. About to book on a different airline and Cancelled Flight my card."
- **Neutral:** "@SouthwestAir I absolutely will. Where can I send it?"

2.6 Conclusion

The dataset used in this study offers a comprehensive view of customer sentiments regarding airline services. Through diagnostic and preprocessing steps, we ensured the dataset's quality and suitability for machine learning models. The analysis highlights key challenges, such as class imbalance and diverse text structures, that are addressed in subsequent stages of this study.

Chapter 3

Results and Discussion

3.1 Data Splitting

The dataset was divided into training and testing subsets using an 75-25 split. The `train_test_split` function from Scikit-learn was utilized, ensuring a fair evaluation of the models.

3.2 Model Training and Performance

Various machine learning models were trained on the dataset, and their performance was evaluated using accuracy scores and classification reports. The results are summarized below:

3.2.1 Support Vector Machine (SVM)

The SVM model achieved an accuracy of 94.00%, demonstrating strong performance in classifying sentiment. The classification report, as shown in Figure 4.1, highlights high precision and recall across all sentiment classes, resulting in an overall F1-score of 0.94.

```
Classification Report of SVM:
----------------------
              precision    recall  f1-score   support

           0       0.97      1.00      0.99      2274
           1       0.89      0.95      0.92      2258
           2       0.97      0.87      0.92      2338

    accuracy                           0.94      6870
   macro avg       0.94      0.94      0.94      6870
weighted avg       0.94      0.94      0.94      6870
```

Figure 3.1: Classification report for the SVM model

To further analyze the performance of the SVM model, we visualized the ROC curve and the confusion matrix. These visualizations, displayed in Figure 4.2, provide additional insights into the model's ability to distinguish between classes and its misclassification rates.

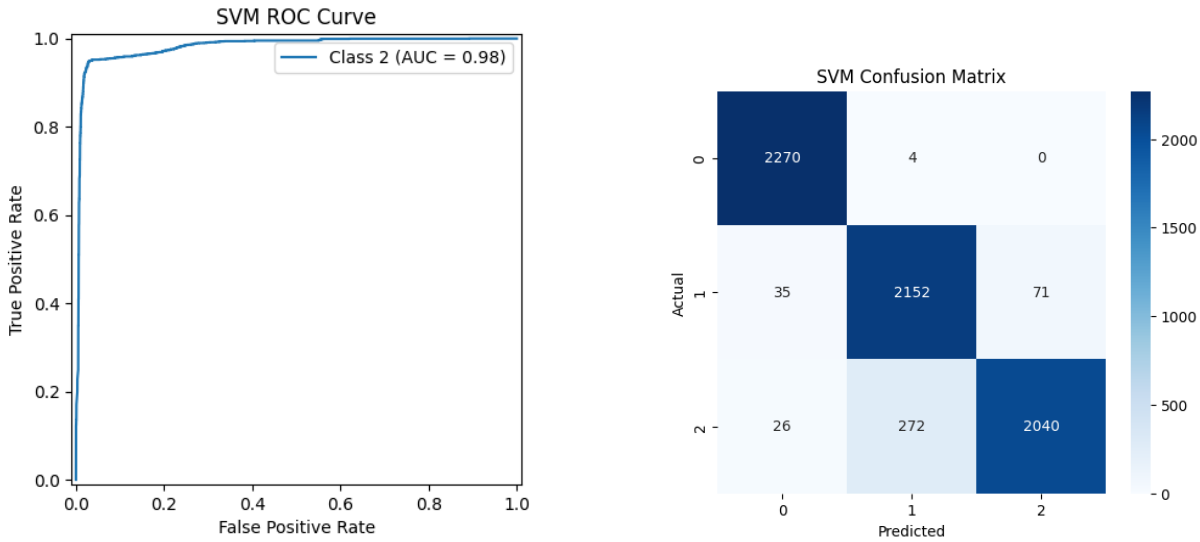

Figure 3.2: Visualization of the SVM model's ROC curve and confusion matrix

3.2.2 KNN

The KNN model, with k=5, achieved a lower accuracy of 59.00%, indicating some difficulty in classifying sentiment accurately. The classification report, as shown in Figure 4.3, reveals challenges in distinguishing between sentiment classes, with a macro-average F1-score significantly lower than that of the SVM model.

```
Classification Report of KNN:
----------------------
              precision    recall  f1-score   support

           0       1.00      0.05      0.10      2274
           1       0.51      0.89      0.65      2258
           2       0.71      0.85      0.78      2338

    accuracy                           0.60      6870
   macro avg       0.74      0.60      0.51      6870
weighted avg       0.74      0.60      0.51      6870
```

Figure 3.3: Classification report for the SVM model

To further explore the performance of the KNN model, we visualized the confusion matrix. This visualization, presented in Figure 4.4, offers additional insights into the model's ability to differentiate between classes and its misclassification tendencies.

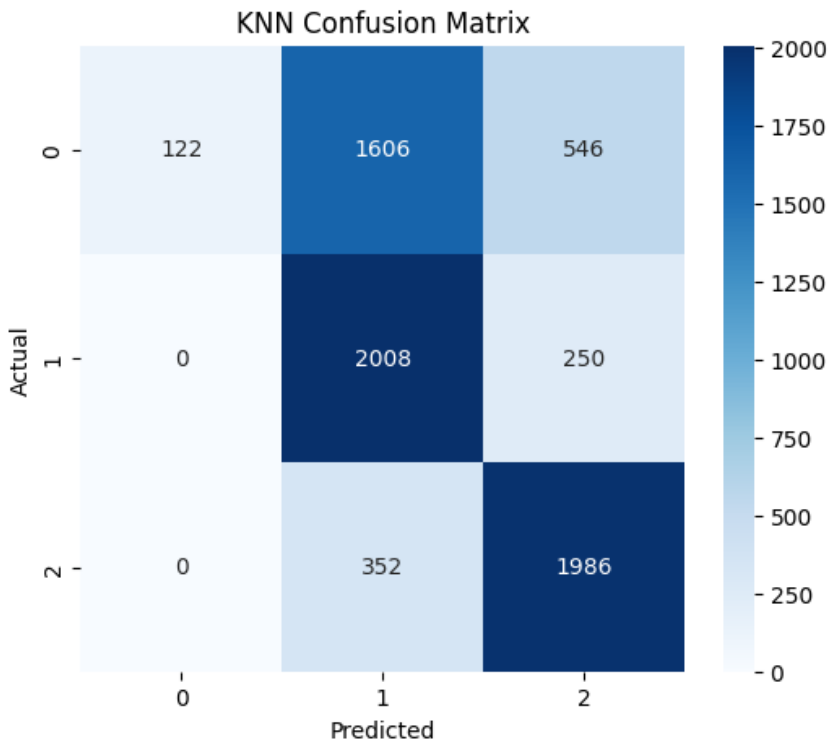

Figure 3.4: Confusion matrix off KNN model

3.2.3 Naive Bayes

The Naive Bayes classifier achieved an accuracy of 86.00%, demonstrating a solid performance in sentiment classification. The classification report, as shown in Figure 4.5, indicates balanced precision and recall across all sentiment classes, resulting in an overall F1-score of 0.86.

```
Classification Report of Naive Bayes:
----------------------
              precision    recall  f1-score   support

           0       0.80      0.97      0.87      2274
           1       0.90      0.72      0.80      2258
           2       0.90      0.89      0.89      2338

    accuracy                           0.86      6870
   macro avg       0.87      0.86      0.86      6870
weighted avg       0.87      0.86      0.86      6870
```

Figure 3.5: Classification report for the Naive Bayes model

To further assess the performance of the Naive Bayes model, we visualized the ROC curve and the confusion matrix. These visualizations, displayed in Figure 4.6, offer deeper insights into the model's classification behavior and the distribution of its errors.

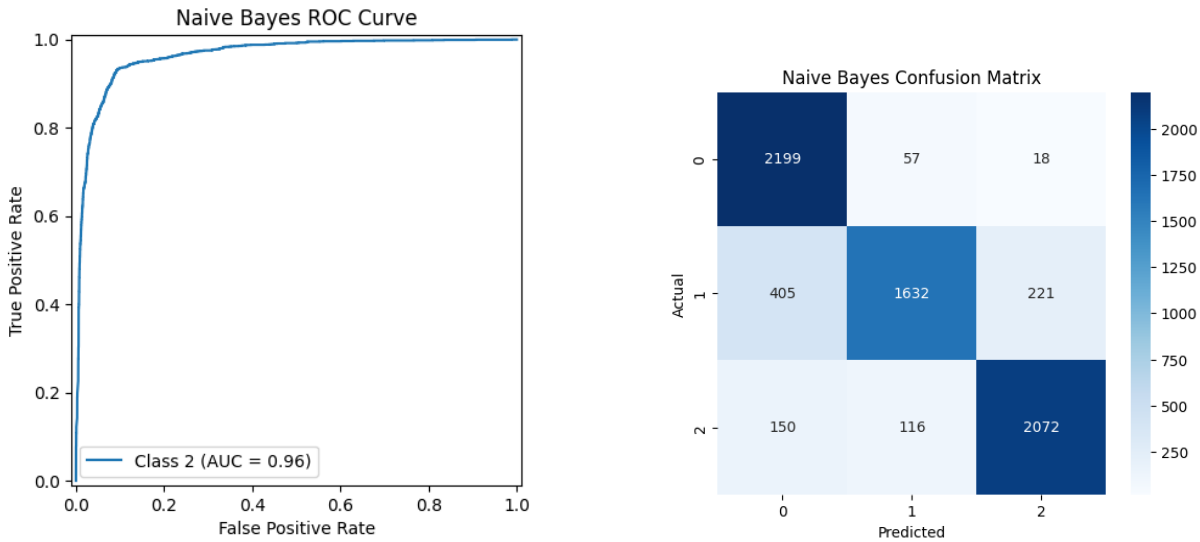

Figure 3.6: Visualization of the Naive Bayes model's ROC curve and confusion matrix

3.2.4 Decision Tree

The Decision Tree model demonstrated strong performance, achieving an accuracy of 94.00%. It exhibited comparable metrics to the SVM model, with high precision and recall across sentiment classes. However, due to the nature of decision trees, it may be prone to overfitting. The classification report, as shown in Figure 4.7, provides further details on the model's performance.

```
Classification Report of Decision Tree:
----------------------
              precision    recall  f1-score   support

           0       0.99      0.98      0.99      2274
           1       0.92      0.91      0.91      2258
           2       0.91      0.93      0.92      2338

    accuracy                           0.94      6870
   macro avg       0.94      0.94      0.94      6870
weighted avg       0.94      0.94      0.94      6870
```

Figure 3.7: Classification report for the Decision Tree model

To further evaluate the performance of the Decision Tree model, we visualized the ROC curve and the confusion matrix. These visualizations, displayed in Figure 4.8, offer additional insights into the model's classification behavior and its misclassification rates.

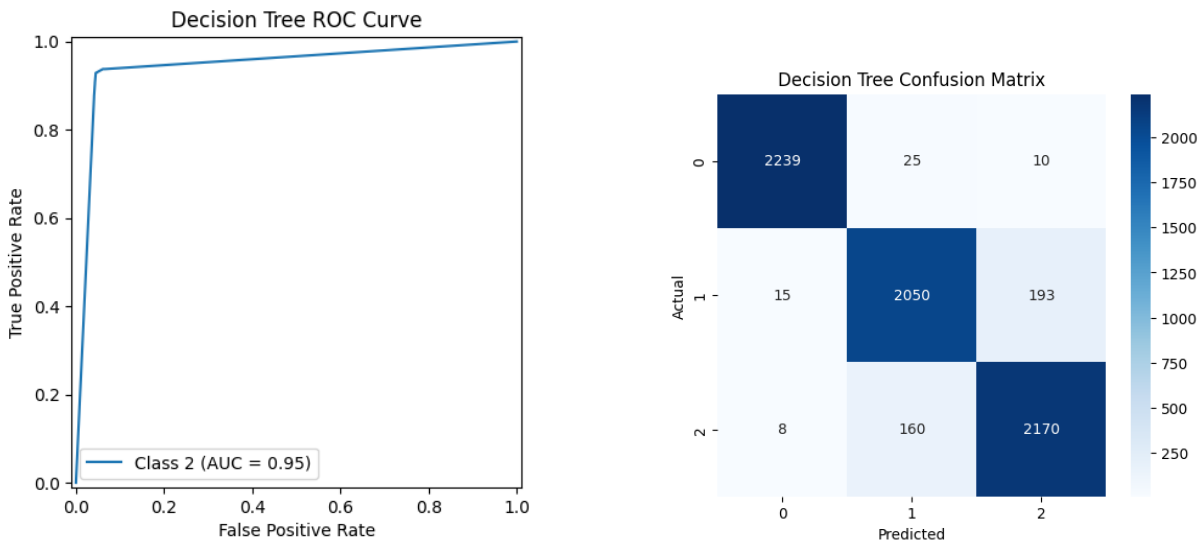

Figure 3.8: Visualization of the Decision Tree model's ROC curve and confusion matrix

3.2.5 Random Forest

The Random Forest classifier achieved the highest accuracy of 96.00%, demonstrating superior performance compared to other models. Its ensemble approach enabled it to outperform all other models in terms of precision, recall, and F1-score across all sentiment classes. The classification report, shown in Figure 4.9, provides detailed insights into its performance.

```
Classification Report of Random Forest:
----------------------
              precision    recall  f1-score   support

           0       0.98      1.00      0.99      2274
           1       0.95      0.94      0.94      2258
           2       0.95      0.94      0.95      2338

    accuracy                           0.96      6870
   macro avg       0.96      0.96      0.96      6870
weighted avg       0.96      0.96      0.96      6870
```

Figure 3.9: Classification report for the Random Forest model

To further analyze the performance of the Random Forest model, we visualized the ROC curve and the confusion matrix. These visualizations, displayed in Figure 4.10, offer a deeper understanding of the model's classification behavior and its misclassification rates.

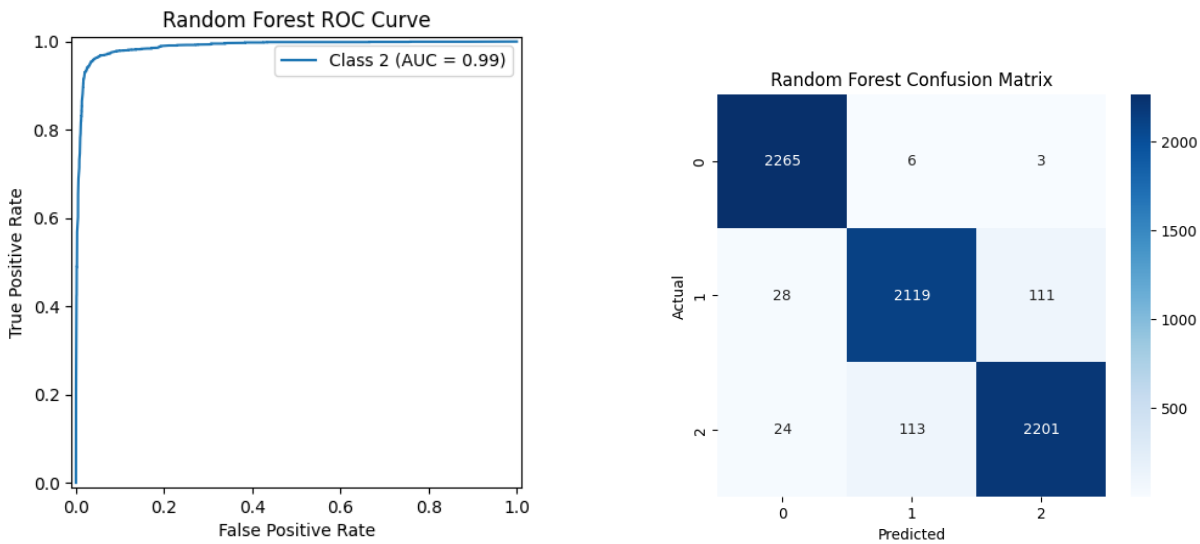

Figure 3.10: Visualization of the Random Forest model's ROC curve and confusion matrix

3.3 Performance Summary

Table 3.1 provides a comparative overview of the model performances:

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
Support Vector Machine	94.00	94.00	94.00	94.00
K-Nearest Neighbors	60.00	74.00	60.00	51.00
Naive Bayes	86.00	87.00	86.00	86.00
Decision Tree	94.00	94.00	94.00	94.00
Random Forest	96.00	96.00	96.00	96.00

Table 3.1: Comparison of different Models performance

3.4 Discussion

The Random Forest model demonstrated the best performance among all classifiers, with the highest accuracy and balanced precision-recall metrics. This is likely due to its ensemble nature, which reduces overfitting and captures complex patterns in the data. The SVM and Decision Tree models also showed competitive results, making them viable options for sentiment analysis tasks.

The KNN model, despite being straightforward to implement, underperformed due to its sensitivity to data scaling and sparsity. Naive Bayes, while efficient, struggled with nuanced sentiment detection, likely due to its assumption of feature independence.

Future work could involve hyperparameter optimization, feature engineering, and exploring advanced deep learning models to further enhance performance.

Conclusion

This study explored the application of machine learning techniques to sentiment analysis using social media data, specifically customer tweets about U.S. airlines. By employing preprocessing strategies, handling class imbalances, and evaluating multiple models, we demonstrated the potential of algorithms like Random Forest and SVM to deliver highly accurate sentiment classification. Among the models tested, Random Forest achieved the highest accuracy (96%) and overall performance, underscoring the advantages of ensemble learning for text-based sentiment analysis.

The findings emphasize the critical role of automated sentiment analysis in bridging the gap between raw textual data and actionable business insights. This approach enables businesses to monitor customer satisfaction trends, address operational inefficiencies, and optimize strategies in real time. Future work can explore deep learning techniques and real-time data processing to further enhance sentiment analysis capabilities.